

Master 17 DSA Patterns in Python

ULTIMATE GUIDE FOR CODING INTERVIEWS

Introduction to DSA Patterns

- ▶ DSA (Data Structures and Algorithms) ke problems ko solve karne ke liye kuch specific patterns hote hain.
- ▶ Agar aap in patterns ko identify karna seekh lete hain, toh coding interviews mein kisi bhi naye question ko solve karna kaafi aasan ho jata hai.
- ▶ Python mein in patterns ko implement karna thoda aasan hota hai kyunki isme in-built libraries (jaise collections.deque, heapq) bahot powerful hoti hain.
- ▶ Yahan sabse important aur commonly puche jane wale 17 DSA Patterns ki list hai, unke concepts aur example questions ke sath:



| 1. Sliding Window

- ▶ **Concept:** Yeh pattern arrays ya strings mein contiguous subarray ya sublist dhoondhne ke liye use hota hai. Isme ek "window" (do pointers se bani hui) hoti hai jo data ke upar slide karti hai.
- ▶ **Python Tips:** Simple for loop aur while loop ka use hota hai window size adjust karne ke liye.
- ▶ **Popular Questions:**
 - Maximum Sum Subarray of Size K
 - Longest Substring Without Repeating Characters
 - Fruits into Baskets

2. Two Pointers

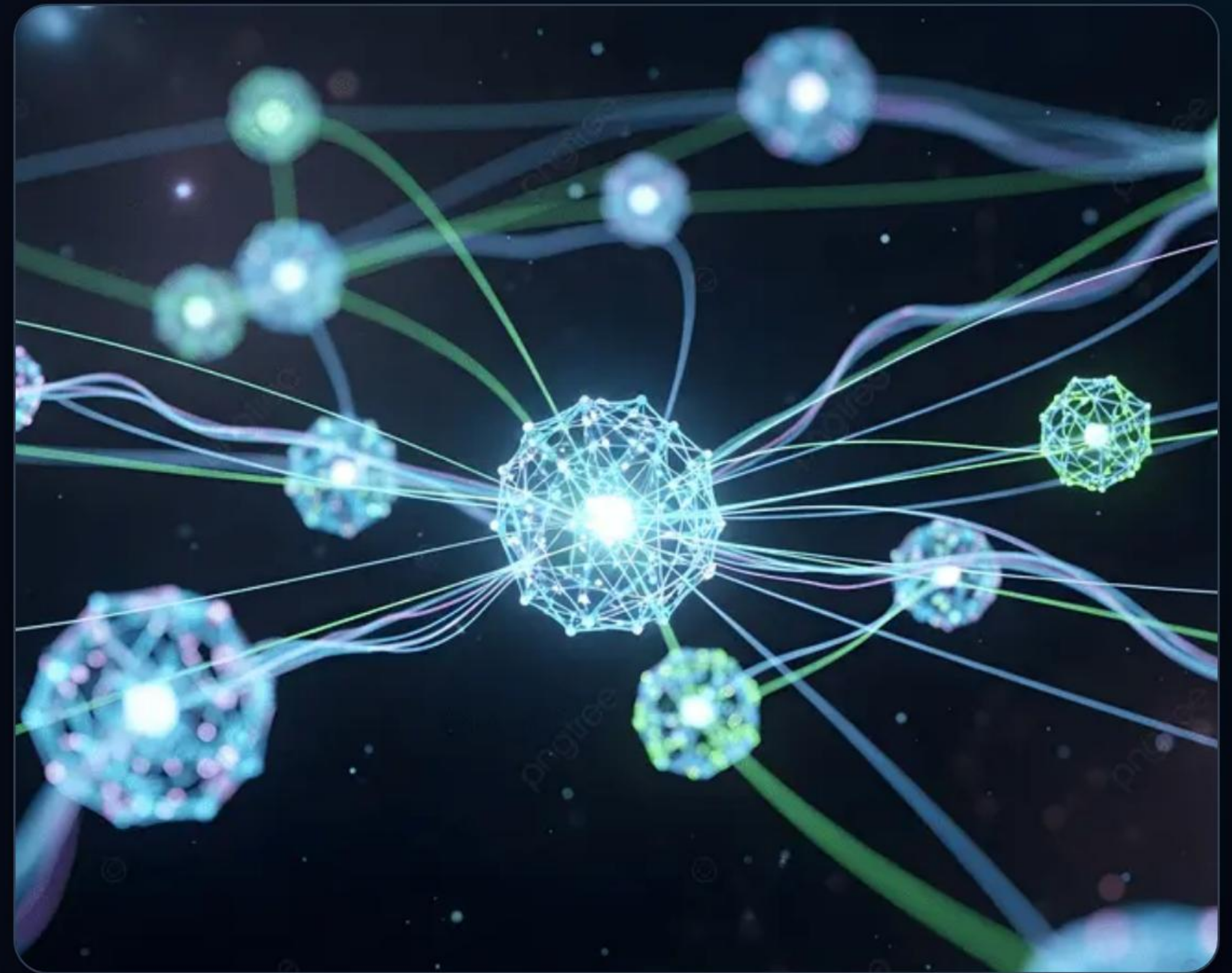
Concept: Sorted arrays ya linked lists mein pairs dhoondhne ke liye. Ek pointer start se aur ek end se move karta hai, ya dono alag-alag speed se move karte hain.

Popular Questions:

- Pair with Target Sum (Two Sum)
- Squaring a Sorted Array
- Remove Duplicates

3. Fast & Slow Pointers (Tortoise & Hare)

- ▶ **Concept:** Yeh pattern LinkedList mein cycle detect karne ya middle node dhoondhne ke liye use hota hai. Ek pointer slow (1 step) aur dusra fast (2 steps) move karta hai.
- ▶ **Popular Questions:**
 - LinkedList Cycle
 - Middle of the LinkedList
 - Happy Number



| 4. Merge Intervals

- ▶ **Concept:** Jab bhi overlapping intervals (ranges) diye ho aur unhe merge karna ho ya overlaps check karne ho.
- ▶ **Python Tips:** Pehle intervals ko unke start time ke hisaab se sort() karna zaroori hota hai.
- ▶ **Popular Questions:**
 - Merge Intervals
 - Insert Interval
 - Meeting Rooms (Conflicting Appointments)

5. Cyclic Sort

Concept: Agar aapko ek array diya hai jisme numbers ek specific range mein hain (jaise 1 se N), toh unhe unki sahi index par place karke missing ya duplicate numbers dhoondhne ka sabse fast tareeqa hai.

Popular Questions:

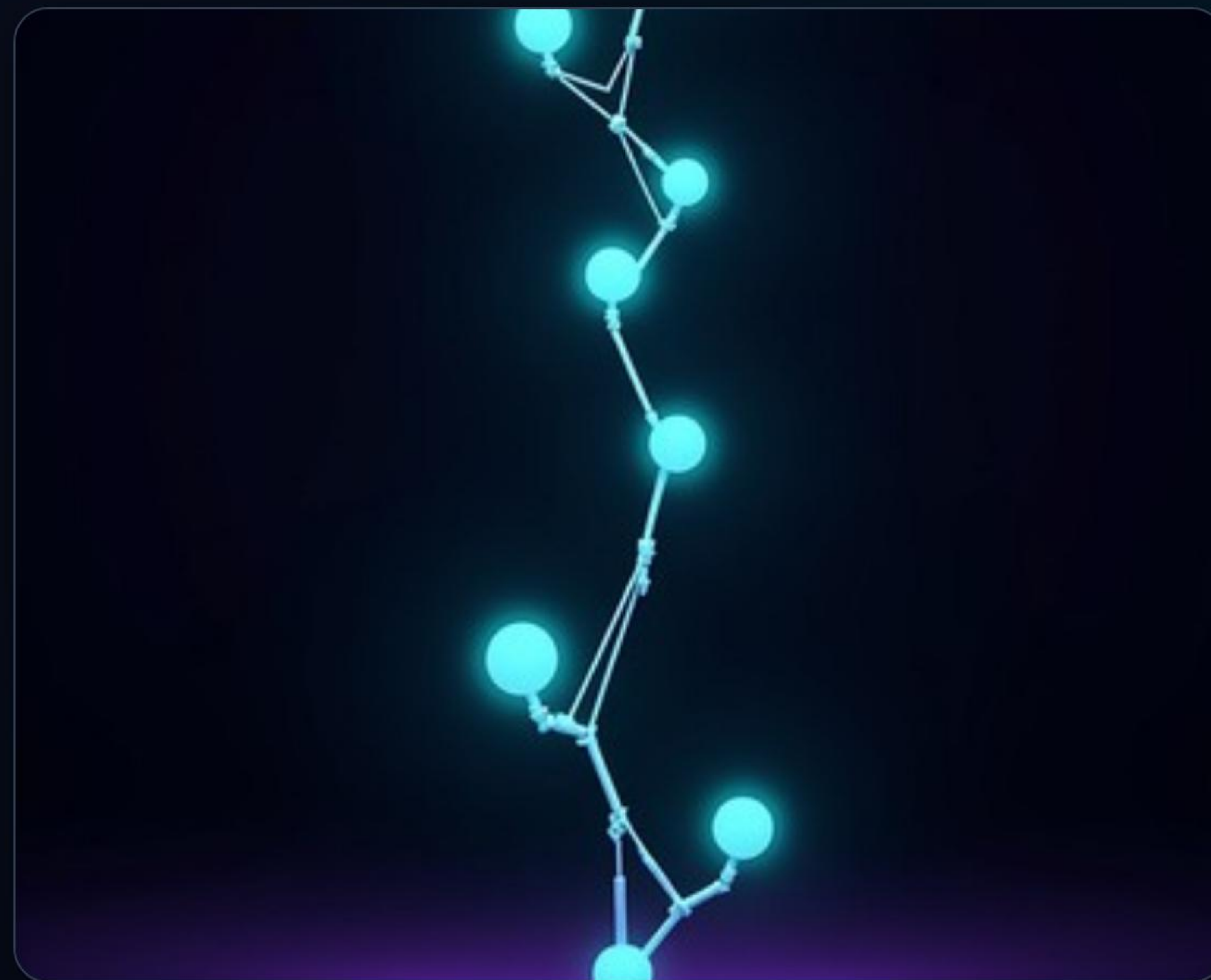
- Find the Missing Number
- Find All Duplicate Numbers

| 6. In-place Reversal of a LinkedList

- ▶ **Concept:** LinkedList ke nodes ke links ko reverse karna bina extra space ($O(1)$ memory) use kiye.
- ▶ **Popular Questions:**
 - Reverse a LinkedList
 - Reverse a Sub-list (Reverse Nodes in k-Group)

7. Tree Breadth First Search (BFS)

- ▶ **Concept:** Tree ya Graph ko level-by-level traverse karna.
- ▶ **Python Tips:** Iske liye hamesha Python ka `collections.deque` use karein (queue ki tarah) taaki pop and push $O(1)$ time mein ho.
- ▶ **Popular Questions:**
 - Binary Tree Level Order Traversal
 - Zigzag Traversal



8. Tree Depth First Search (DFS)

Concept: Tree ke root se leaf tak deep traverse karna. Yeh zyada tar Recursion ya Stack ka use karke hota hai.

Popular Questions:

- Path Sum
- All Paths for a Sum
- Lowest Common Ancestor

| 9. Two Heaps

- ▶ **Concept:** Jab array ko do parts mein divide karna ho jisme ek se smallest element aur dusre se largest element jaldi chahiye ho.
- ▶ **Python Tips:** Python mein heapq module by default Min-Heap hota hai. Max-Heap ke liye values ko negative (-) karke push karna padta hai.
- ▶ **Popular Questions:**
 - Find the Median of a Number Stream
 - Sliding Window Median

| 10. Subsets (Backtracking)

Concept: Combinations aur Permutations find karne ke liye. Yeh recursive aur BFS dono tarah se banaya ja sakta hai.

Popular Questions:

- Subsets (Power Set)
- Permutations
- Combination Sum

11. Modified Binary Search

- ▶ **Concept:** Normal binary search ko modify karke rotated sorted arrays ya order-agnostic arrays pe lagana.
- ▶ **Popular Questions:**
 - Search in a Rotated Array
 - Find Minimum in Rotated Sorted Array



| 12. Top 'K' Elements

- ▶ **Concept:** Array mein se Top/Smallest/Largest/Most Frequent 'K' elements find karna.
- ▶ **Python Tips:** Isme hamesha heapq (Heap) use hota hai jo $O(N \log K)$ time complexity deta hai.
- ▶ **Popular Questions:**
 - Top K Frequent Elements
 - Kth Largest Element in an Array

13. K-way Merge

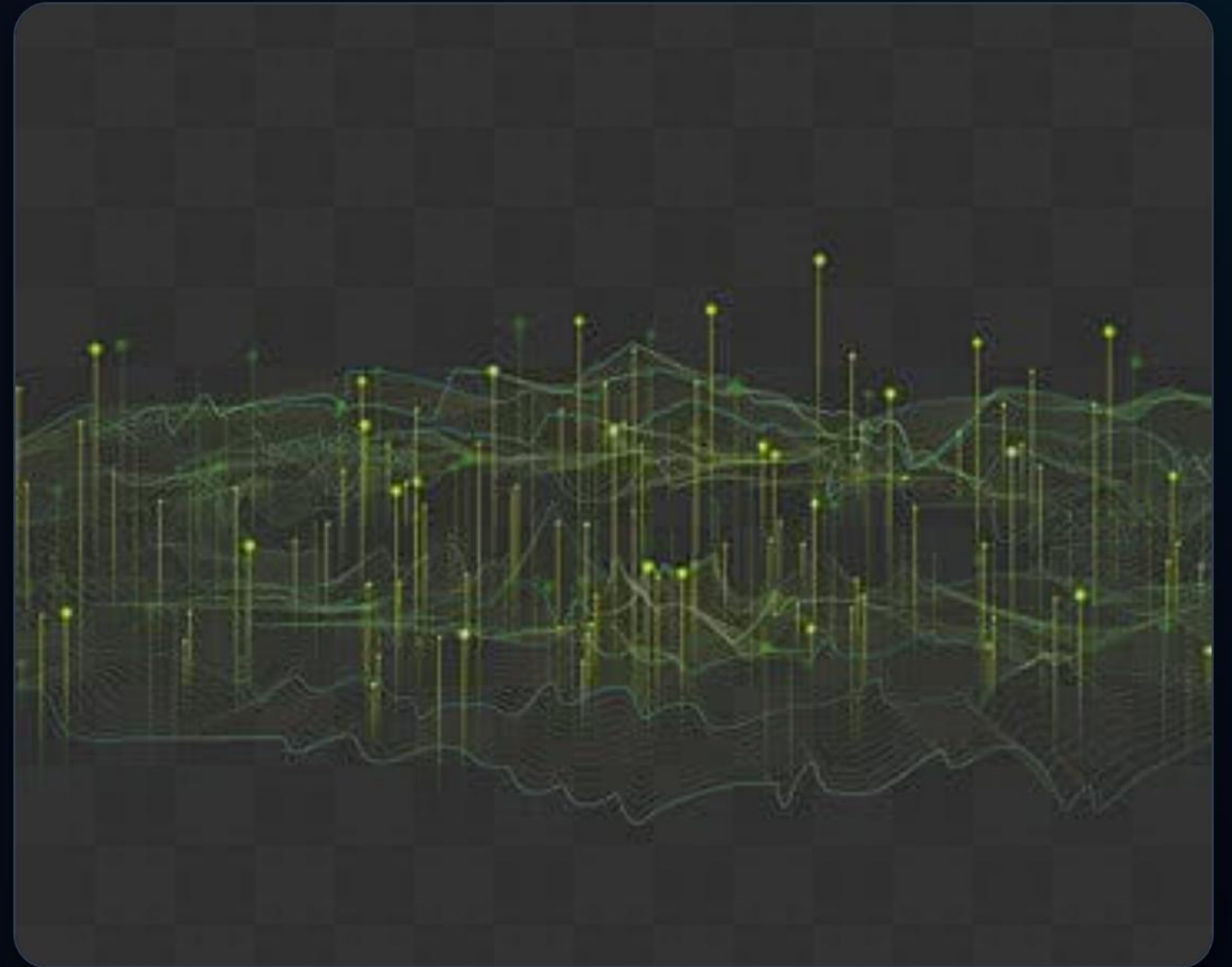
Concept: K sorted arrays ya linked lists ko ek single sorted list mein merge karna. Yeh bhi Heaps ka use karta hai.

Popular Questions:

- Merge K Sorted Lists
- Kth Smallest Number in M Sorted Lists

14. 0/1 Knapsack (Dynamic Programming)

- ▶ **Concept:** Optimization problems jahan aapko select karna hota hai ki kisi item ko include karna hai ya exclude karna hai (take it or leave it) maximum profit ke liye.
- ▶ **Popular Questions:**
 - 0/1 Knapsack
 - Equal Subset Sum Partition
 - Coin Change



| 15. Topological Sort (Graphs)

- ▶ **Concept:** Directed Acyclic Graphs (DAG) mein dependencies ko resolve karna. Jaise, task B ko task A ke baad hi karna hai.
- ▶ **Python Tips:** Isme Indegree array aur BFS (Kahn's Algorithm) ka use hota hai.
- ▶ **Popular Questions:**
 - Topological Sort
 - Course Schedule (Task Scheduling)

16. Trie (Prefix Tree)

Concept: Strings aur dictionary se related fast searches (jaise autocomplete feature) ke liye ek special tree data structure.

Popular Questions:

- Implement Trie (Prefix Tree)
- Word Search II

17. Monotonic Stack / Queue

- ▶ **Concept:** Ek aisi stack ya queue jisme elements ya toh strictly increasing order mein hote hain ya strictly decreasing order mein.
- ▶ **Popular Questions:**
 - Next Greater Element
 - Daily Temperatures
 - Largest Rectangle in Histogram



Conclusion

Yeh patterns kaafi comprehensive hain aur zyada tar interviews inhi ke aas-paas ghoomte hain.

Kya aap inme se kisi specific pattern ka Python mein step-by-step code aur explanation dekhna chahenge?

Image Sources



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